

0.1 RQ6-Specific Coding Codebook

Coding purpose. RQ6 examines how the included reviews treat the shared infrastructure on which reproducible research depends, whether they audit that infrastructure critically, and which recurring infrastructure gaps they identify.

What the scale measures. The shared 0–3 scale records treatment intensity, meaning how prominently a review treats a coded infrastructure element, critique, or gap within its own account of the literature. It does not measure whether the primary studies were in fact reproducible, and it does not measure the quality of any simulator, dataset, or benchmark. A score of 0 on code availability means the review does not address code availability, not that code is unavailable. The score describes the review, not the field.

Table 1: Shared treatment-intensity scale as applied in RQ6.

Score	Label	General interpretation
0	Not addressed	The element, critique, or gap is not identifiable in the review.
1	Mentioned	It is named or noted in passing, without development.
2	Discussed	It is explained and used in the review’s discussion, but does not organize part of the synthesis.
3	Structuring role	It organizes part of the review’s synthesis, through a taxonomy, an explicit comparative criterion, a dedicated section, or a recurring interpretive theme.

Structure of RQ6 coding. RQ6 records three survey-level scores and three unit-level tables.

Reproducibility auditing records how far the review examines whether reviewed work could be reproduced, as distinct from describing the infrastructure used.

Infrastructure auditing records how far the review documents the simulators, datasets, benchmarks, artifacts, and reporting practice of the studies it covers.

Blind-spot visibility records how far the review makes visible what the literature systematically fails to provide.

Infrastructure coverage has six units recording how visibly the review treats each element: simulators or environments; datasets or data sources; benchmark tasks; code or artifact availability; external signals; and reporting or experimental detail.

Critical audit has five units recording how far the review criticizes infrastructure practice rather than only describing it: benchmark comparability critique; open-code or reproducibility critique; simulator fragmentation critique; data realism or availability critique; and the need for standardization or shared infrastructure.

Recurring infrastructure gaps has five units: closed code or unavailable artifacts; absence of shared benchmarks; inconsistent external signals or assumptions; simulator or testbed fragmentation; and weak dataset realism or documentation.

Coverage against critique. The coverage and critical-audit tables are deliberately separate. A review can document which simulators the literature uses without arguing that simulator fragmentation is a problem. Coverage records visibility. Critical audit records evaluative stance. A unit may therefore score highly on coverage and zero on critique.

Relationship between survey-level and unit-level scores. All three survey-level scores are coded directly from the review and are not derived from unit scores. Infrastructure auditing coincides with the highest coverage unit in most reviews, and blind-spot visibility coincides with the highest gap unit in most reviews, but neither relationship holds throughout, so neither is an aggregation. Reproducibility auditing does not track any single unit.

Zero and applicability. A score of 0 means the review does not address the element, critique, or gap. The scale does not separate a topic that a review considered and did not develop from one that fell outside its declared scope. Both are recorded as 0. This keeps denominators uniform and is consistent with the construct, which records the treatment a review gives a topic rather than the reasons for that treatment. A score of 0 should therefore not be read as evidence that a review overlooked something it could have covered.

All coded reviews are scored on every dimension and unit. RQ6 has no applicability route, so the denominator is the full coded corpus.

Table 2: Operational criteria for infrastructure-coverage units.

Score	Criterion
0	The element is not identifiable in the review.
1	The element is named in passing, for example a single remark that studies use different simulators.
2	The element is described substantively, with examples or an account of what reviewed studies use.
3	The element is documented systematically across reviewed studies, for example in a comparative table, or organizes part of the synthesis.

Operational criteria: infrastructure coverage.

Table 3: Operational criteria for critical-audit units.

Score	Criterion
0	The review does not take an evaluative position on this aspect of infrastructure practice.
1	A concern is registered without elaboration.
2	The review explains why the practice is problematic and what it costs research.
3	The critique is a structuring concern, developed as a recurring theme, a dedicated discussion, or an explicit basis for the review’s recommendations.

Operational criteria: critical audit.

Table 4: Operational criteria for infrastructure-gap units and for the survey-level blind-spot score.

Score	Criterion
0	The gap is not identified.
1	The gap is acknowledged as a limitation without elaboration.
2	The gap is explained, with its consequence for cumulative research made clear.
3	The gap is a structuring concern, developed as a recurring theme, a dedicated discussion, or an explicit basis for the review’s recommendations.

Operational criteria: recurring infrastructure gaps.

Operational criteria: reproducibility auditing.

Evidence rules. Tables, figures, taxonomies, and dedicated sections are coding evidence, and organizing use of an element, critique, or gap in any of them supports a structuring score. Naming the simulator used by a reviewed study, without comparison or comment, supports a

Table 5: Operational criteria for the survey-level reproducibility-auditing score.

Score	Criterion
0	Whether reviewed work could be reproduced is not considered.
1	Reproducibility is mentioned, for example through a remark on limited reporting or unavailable material, without development.
2	The review discusses reproducibility substantively, but does not organize any part of its synthesis around it.
3	Reproducibility organizes part of the review, for example through a dedicated assessment of artifact availability, reporting completeness, or the conditions needed for independent replication.

score of 1. Repeated mentions do not accumulate into a higher score. Content appearing in the abstract or keywords alone is scored at most 1. Supplementary material was not coded, because its availability is inconsistent across the corpus and coding it would make reviews non-comparable. Coding reflects the review authors’ own account: a code repository linked by a cited primary study is insufficient unless the review discusses, tabulates, or evaluates artifact availability. Where evidence falls between two adjacent levels, the lower score is recorded unless the higher criterion is clearly satisfied.

Worked boundary examples. The examples below are taken from the survey-level audit tables that follow.

Table 6: Worked boundary examples for RQ6 coding.

Review	Dimension or unit	Score	Basis for the boundary
Mason and Grijalva (2019)	Reproducibility auditing, survey	1	The review is infrastructure-aware, treating simulators and external signals, but does not examine whether reviewed work could be reproduced. Boundary against 2.
Wang and Hong (2020)	Reproducibility auditing, survey	3	Reproducibility organizes part of the review rather than appearing as an incidental concern. Contrast with the preceding row.
Mason and Grijalva (2019)	Code availability, coverage	0	Artifact availability is not addressed. Boundary against 1, which would require at least a passing remark.
Mason and Grijalva (2019)	Simulator fragmentation, critique	1	A concern about differing simulators is registered without explaining what it costs cumulative research. Boundary against 2.
Shaqour and Hagishima (2022)	Open-code critique	3	Absent open code is developed as a recurring basis for the review’s conclusions rather than noted once.
Wang and Hong (2020)	Simulators, coverage vs critique	3 and 2	Simulators are documented systematically, which is a coverage score of 3, while the review explains what fragmentation costs research without organizing the synthesis around it, which is a critique score of 2. Illustrates that coverage and critique are independent.

Summary fields. Each survey-level audit table also carries three narrative fields: reproducibility orientation, dominant infrastructure concern, and the main RQ6 consequence. These summarize the coded scores in prose and are not themselves coded values.